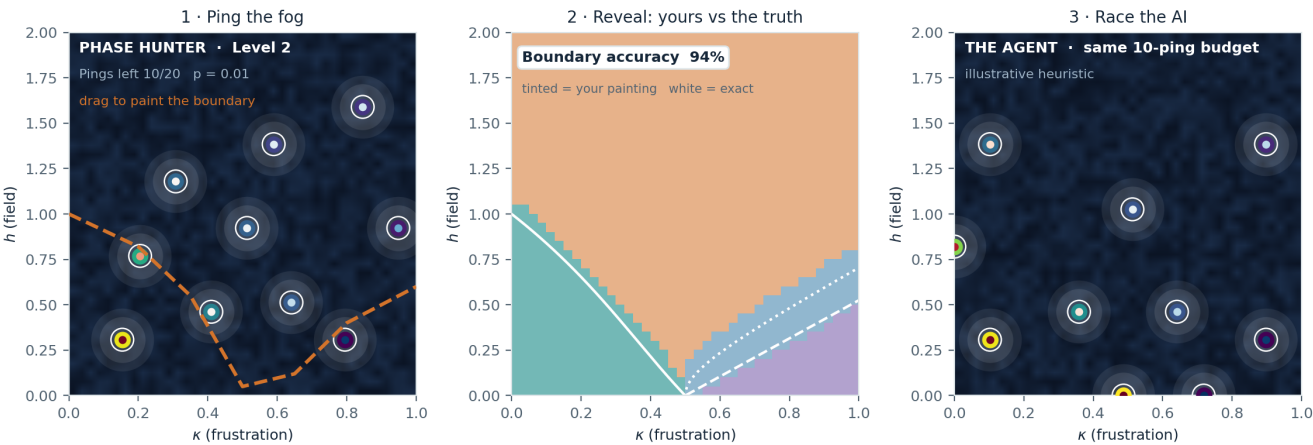


Phase Hunter

A Battleship-style game on a hidden quantum phase diagram, with the real science underneath.

The hook. You get 20 measurements. Somewhere in the fog are four phases of a quantum magnet. Find the boundaries before the AI does, then watch them blur as quantum noise rises.



Mock-up of the three main screens (ping the fog, reveal against the truth, watch the agent). Ping colours come from real 8-spin simulation data with real shot noise; the interface itself is not built yet.

Track / format	Scientific Track (Open Challenge). Required: notebook, 3 phase diagrams ($p = 0, 0.01, 0.05$), 2-3 page writeup, 5-7 minute presentation.
Deadlines (from the event schedule)	Draft: Wed Sept 23, 11:59 PM. Final: Fri Sept 25, 11:59 PM. Confirm the timezone on the dashboard.
Who builds it	Tanish (owner: decisions, physics review, presentation) and Claude (builds simulation, game, agent, figures). Teammates: playtest, review, rehearse.
Status of this document	A proposal. Figures marked <i>real</i> use exact-diagonalization data computed for this plan (8 spins). Nothing here is a final result.

THE IDEA

One page: what we are building and why

A game where the fun and the science score are the same thing.

What it is. The hidden map is the phase diagram of the 1D ANNNI spin chain in the (κ, h) plane, the exact object the Scientific Track asks us to map. A player spends a limited budget of **pings** (simulated measurements at chosen (κ, h) points), reads the results, and **paints the phase boundaries**. The game scores the painting against the reference boundaries from the starter kit, then replays what an **AI agent** did with the same budget.

Why a game at all. Judges see many notebooks with heatmaps. A judge who plays a level against our agent for two minutes will remember it, and while playing they feel the hardest part of the physics: measurements are noisy, and noise on the gates blurs the phases. But the score comes from the science, so the game is built on top of the science and never replaces it.

Rubric line (from the handout)	Weight	What Phase Hunter contributes	Evidence we will produce
Phase diagram accuracy	25%	Clean and noisy diagrams from a real simulation pipeline, scored against the analytic lines with the same metric the game uses	3 diagrams + cell accuracy vs reference
Noise analysis depth	25%	Each noise level is a game level; boundary shifts are measured per phase, not described loosely	Table: boundary position at fixed κ for $p = 0, 0.01, 0.05$; which phase shrinks first
Physical insight	20%	Levels are built around why noise hurts (long-range order, $\uparrow\uparrow\downarrow\downarrow$ pattern) and why the floating phase is hard	Writeup section + boss level
Technical sophistication	15%	Simulation with depolarizing noise, an uncertainty-driven agent, a measurement-budget study, optional error mitigation	Pings-vs-accuracy study; optional extrapolation to $p = 0$
Presentation and clarity	15%	A judge plays live against the agent	5-7 minute demo script (page 9)

Guardrail. The required deliverables come first: notebook, three phase diagrams, writeup, presentation. If time runs short we cut game features, never science. The cut order is on page 8.

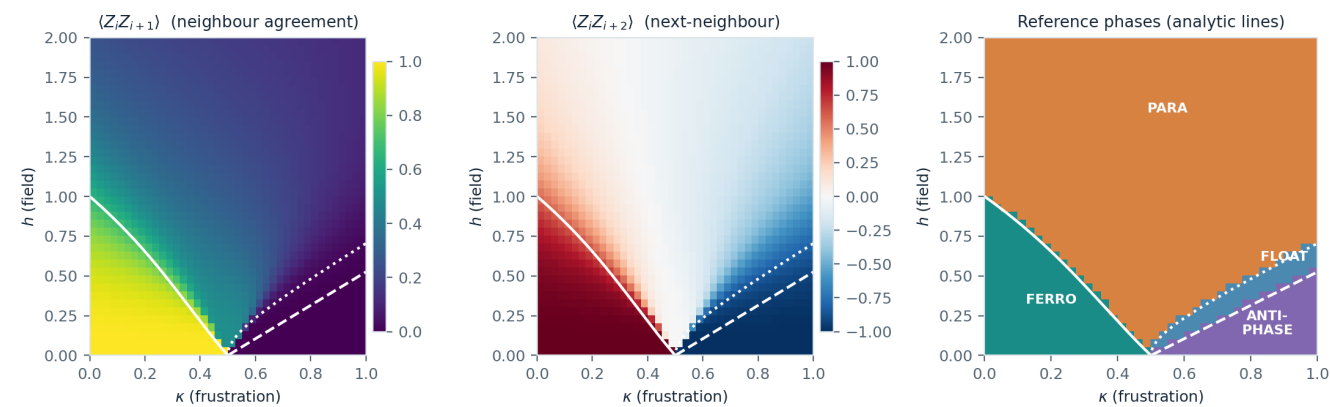
What is real and what is not. The game runs in a browser, where a live noisy quantum simulation is too slow. So we precompute the measurement statistics for a grid of (κ, h) points with PennyLane and exact diagonalization, and the game samples bitstrings from those stored distributions, with real shot noise. That is real simulation data, and the writeup will say exactly how it was produced. The game does not run a quantum computer live, and we will not claim it does.

The ANNNI model and its four phases

Everyone on the team should be able to say this in their own words.

A row of spins, each pointing up or down (and able to be in superposition). The Hamiltonian is $H = -\sum Z_i Z_{i+1} + \kappa \sum Z_i Z_{i+2} - h \sum X_i$. The first term wants neighbours to agree. The second (frustration, strength κ) wants spins two apart to *disagree*. The third (a sideways field, strength h) wants every spin in superposition. Which term wins in the ground state decides the phase.

Phase	Where (κ , h)	What the spins do	What we measure
Ferromagnetic	$\kappa < 0.5$, low h	All agree: $\uparrow\uparrow\uparrow\uparrow$ or $\downarrow\downarrow\downarrow\downarrow$	Neighbour correlation $\langle Z_i Z_{i+1} \rangle$ near +1
Antiphase	$\kappa > 0.5$, low h	Repeating $\uparrow\uparrow\downarrow\downarrow\uparrow\uparrow\downarrow\downarrow$ pattern	$\langle Z_i Z_{i+1} \rangle$ near 0, $\langle Z_i Z_{i+2} \rangle$ near -1
Paramagnetic	High h	Field wins; spins mostly in superposition	All correlations small
Floating	Narrow sliver, $\kappa > 0.5$, between antiphase and para	Incommensurate, slowly decaying correlations	Very hard to see at small N (the handout says so)



Real data. Exact ground states of 8 spins on a ring (periodic), on a 40×40 grid. Left and centre: two correlators the game will show as pings. Right: reference phases from the starter kit's analytic lines (solid: Ising, dashed: BKT, dotted: KT). The lines are qualitative at finite size, as the handout warns, and they visibly track the data.

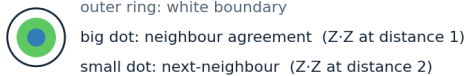
Reference boundaries (from the starter kit). Ising transition for $\kappa < 0.5$; for $\kappa > 0.5$ the antiphase-to-floating line $h = 1.05(\kappa - 0.5)$ and the floating-to-para line $h = 1.05 \cdot \sqrt{(\kappa - 0.5)(\kappa - 0.1)}$. At $\kappa = 1$ those are $h = 0.525$ and $h = 0.704$, so the floating phase is a thin band, which is why it makes a good boss level.

Gotcha we found in the starter kit. `ising_transition( $\kappa$ )` returns 0 at exactly $\kappa = 0$ (the true value is 1, and it is fine at $\kappa = 1e-4$). Any grid that starts at $\kappa = 0$ must guard for this, or the Ising line will look wrong at the left edge. Our figures already do.

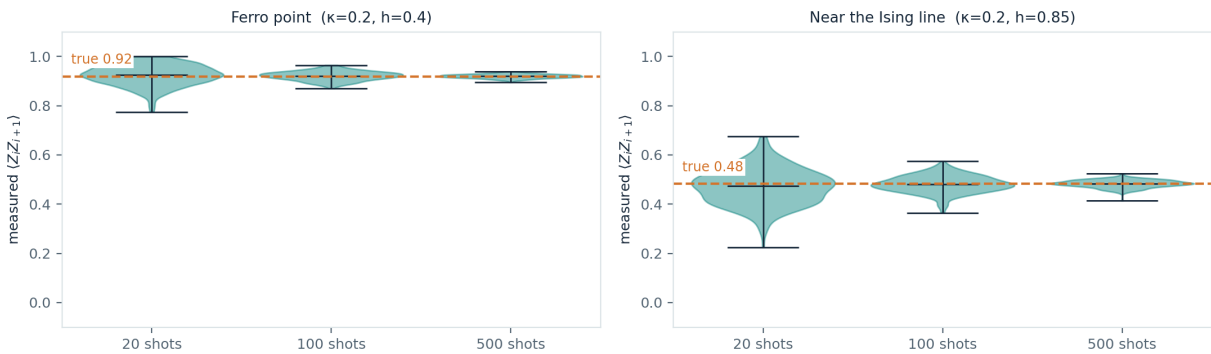
How a round plays

Short rounds, one new decision per level, and an immediate reason to try again.

Step	What the player does	Design intent
1 Pick a level	Choose noise level p and a map seed. Rounds are about 60-90 seconds.	Short loop: easy to say 'one more try'.
2 Ping	Click a (κ, h) point and choose how many shots (measurements) to spend: 20, 100 or 500. Budget: 60 energy; 20 shots cost 1, 100 cost 2, 500 cost 4 (starting values, to be tuned).	The Battleship decision: many cheap noisy pings or a few expensive sharp ones.
3 Read the glyph	Each ping shows a sonar glyph: the big dot is neighbour agreement, the small dot is next-neighbour agreement, both from real sampled bitstrings.	Teaches the physics signatures without a lecture.
4 Paint	Drag handles to place the three boundary curves (Ising; antiphase/floating; floating/para).	The player commits to a map, like calling a shot.
5 Reveal	The exact lines appear over the painting. Score = share of grid cells labelled correctly (the same metric as our science deliverable), with a star rating.	One honest number; shows exactly where they were wrong.
6 Rematch	Replay the agent's pings and painting on the same seed, then instant retry.	'I see what I missed' is the addictive moment.



Glyph legend (drawn once, shown in the tutorial level).



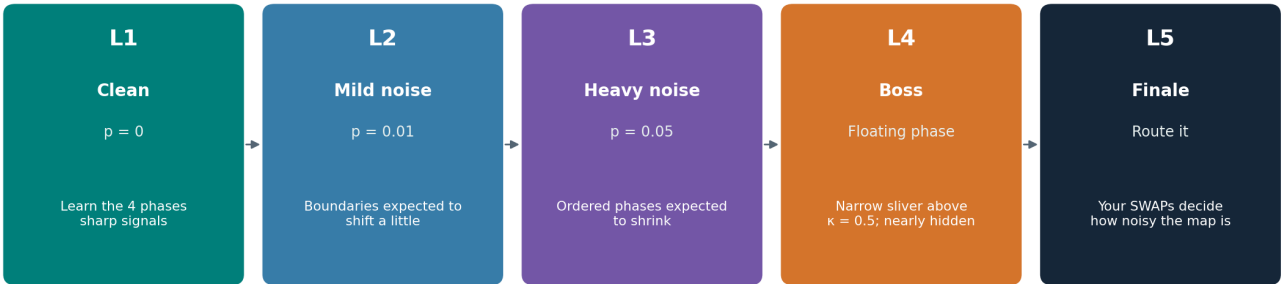
Real data. Why pings mislead: 300 repeated pings of the same point at 20, 100 and 500 shots. Near a phase line (right) 20 shots can read anywhere from about 0.2 to 0.7 when the truth is 0.48. This is the spread the player has to reason about, and the reason the shot-choice decision is real.

Why it is replayable. Every loss has a visible cause (a noisy ping you trusted, a region you never probed). Maps are seeded so friends can compare, and a share card reads like 'Level 3, 94% accuracy, 12 energy'. The numbers in the mock-up are illustrative, not measured player results.

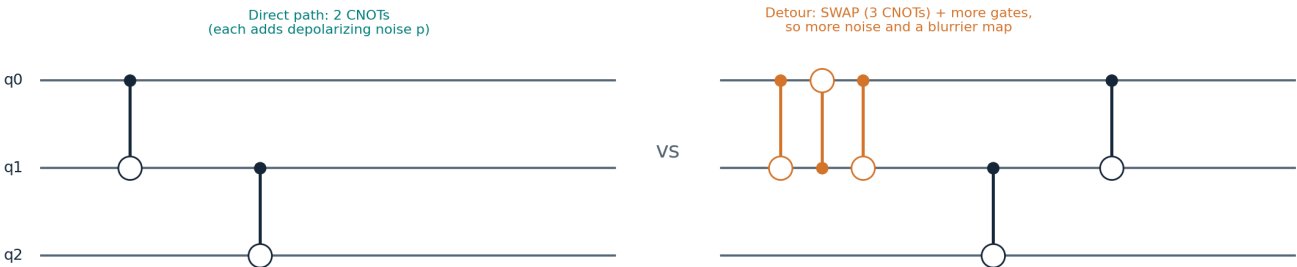
LEVELS

Five levels; each one is a piece of the noise analysis

The level ladder doubles as the outline of the writeup and the presentation.



Level	Noise / twist	What the player learns	Data behind it
L1 Clean	$p = 0$; strong hints	The four phases and their signatures	Exact diagonalization, $N = 8$
L2 Mild noise	$p = 0.01$ per CNOT	Boundaries should shift slightly (per the handout, to be measured); pings are less trustworthy	Noisy-circuit simulation, $N = 6$
L3 Heavy noise	$p = 0.05$ per CNOT	Ordered phases should shrink toward the paramagnet (expected, to be measured)	Noisy-circuit simulation, $N = 6$
L4 Boss: floating phase	Reference band is thin	Why this phase is hard to see at small size, and how measurement budget matters	Same data; scored against the reference band; we report honestly whether we resolve it
L5 Finale: route it (stretch)	Your routing sets the noise	A SWAP costs 3 CNOTs, so a bad route means a blurrier map	Maps at effective noise levels; cut first if time is short

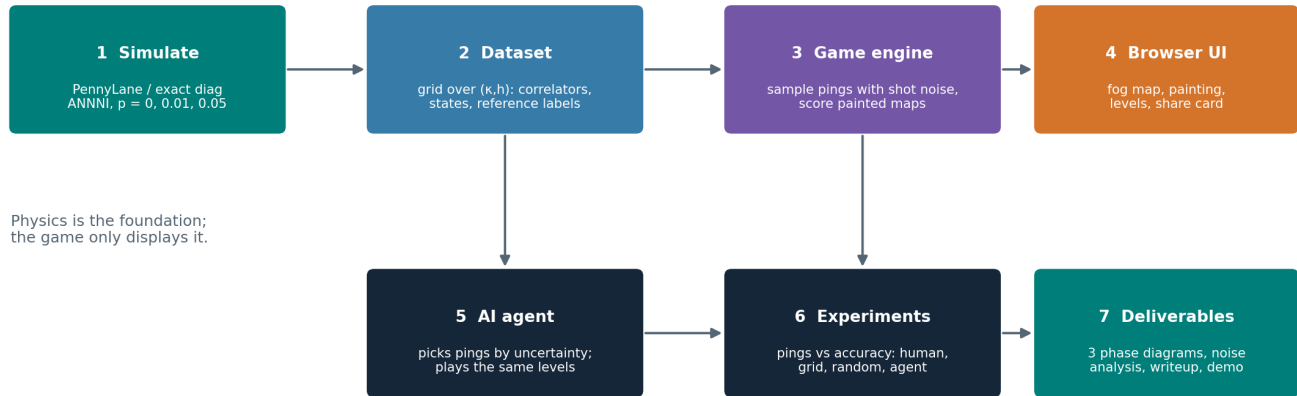


Idea behind the finale: extra SWAPs mean extra CNOTs, and each CNOT adds depolarizing noise. It links the game to the Computational Track's routing problem and shows why compilation matters for physics. Schematic only.

The noise model follows the handout: after each CNOT, the target qubit suffers a random Pauli error with total probability p (X, Y, Z each with $p/3$), simulated with PennyLane's default.mixed device. The levels use $p = 0, 0.01$ and 0.05 , the three values the deliverable requires.

Pipeline: simulate, store, play, analyse

The science pipeline comes first; the game only reads its output.



Seven components. Steps 1-2 and 5-7 carry the required science deliverables; steps 3-4 are the game.

Component	Plan	Notes / open items
1 Simulation, clean	Exact diagonalization, 8 spins, periodic ring, 40×40 grid over k in $[0,1]$, h in $[0,2]$. A numpy prototype already produced this grid in about 10 seconds for this plan.	Verify against PennyLane's own ground state at a few points. Optional: N up to 12 (bonus in the handout).
1 Simulation, noisy	Prepare each ground state with a circuit (ansatz parameters fitted to the exact state), then run that circuit with depolarizing noise after every CNOT on default.mixed. $N = 6$ to keep density matrices small. Grid 20×20 at $p = 0.01$ and 0.05 .	Runtime is not measured yet; first task Monday is to time one point and set N and grid size. Extension: re-optimize parameters under noise.
2 Dataset	Per grid cell and noise level: the full Z-basis probability distribution, plus reference label. Stored as compressed arrays, exported to JSON for the browser.	Storing whole distributions lets the game sample real bitstrings.
Phase classifier (for the required diagrams)	Features: $\langle ZZ \rangle$ at distance 1 and 2, and the antiphase string correlator. Cluster the cells, anchored by three well-known corner points; compare with the analytic labels. Optional second method: QCNN, for the 'multiple methods' bonus.	We report accuracy against the reference and do not tune to it silently.
3-4 Game	Single-file HTML/JavaScript (canvas), no server: fog map, ping sampler, painting handles, scoring, levels, replay, share card.	Can be published as a shareable page for judges.
5 Agent	See next page.	Runs offline in Python; its ping sequences are exported to the game for replay.

Optional error mitigation (bonus). With results at $p = 0.01$ and $p = 0.05$ we can extrapolate each correlator back toward $p = 0$ and check whether the extrapolated map is closer to the clean one. This is a cheap, honest test of one of the handout's bonus goals.

How many measurements does it take to find a phase boundary under noise?

The research question that gives the game a scientific result, not just a demo.

Question. At each noise level, how does boundary accuracy grow with measurement budget, and which strategy for choosing where to ping is best? This is our own experiment on top of the required deliverables; we do not know the answer yet and will report whatever we find, including if the agent does not beat the simple baselines.

Strategy	How it chooses pings	Role in the study
Random	Uniform random points	Baseline
Grid	Evenly spaced points	Baseline players naturally imitate
Human	Playtesters using the game	Small sample; reported as indicative only
Uncertainty agent	Keeps a running phase estimate from the pings so far, then pings where neighbouring results disagree most and coverage is thin	Our proposed method; also the on-stage rival

Design element	Choice
Factors	Strategy x noise level ($p = 0, 0.01, 0.05$) x budget (5 to 40 pings) x shots per ping
Metric	Cell accuracy against the reference labels; also boundary position error along fixed cuts
Repeats	Many random seeds per setting, with spread reported (not just averages)
Output figure	Accuracy vs number of pings, one panel per noise level, one line per strategy
Hypotheses (to test, not claims)	(1) Adaptive pinging needs fewer pings than a grid. (2) Higher noise raises the budget needed. (3) The floating boundary stays hard for every strategy at this system size.

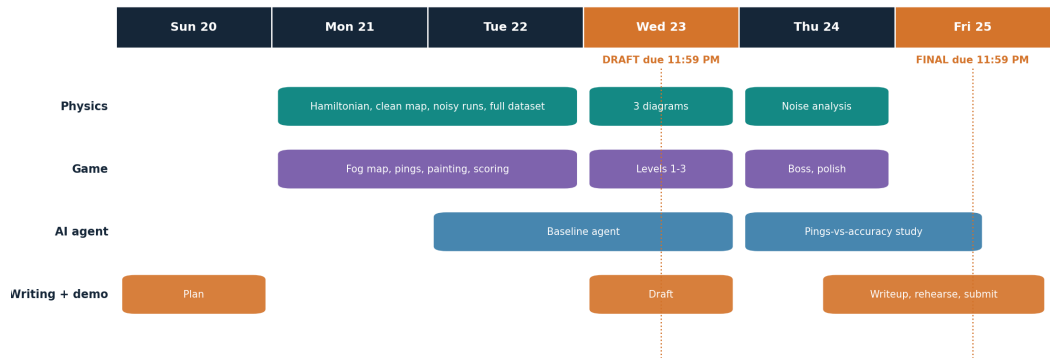
Honesty rule for the writeup and slides. Any statement like 'the agent needs X% fewer pings' goes in only after we have measured it, with the number of repeats and the spread. Until then the slides say 'we tested whether'.

Why this scores. It compares several methods on the same system (a listed bonus), it gives the noise analysis a quantitative backbone (which phase needs more budget, how much), and it is novel enough to lift the technical-sophistication line without heavy new physics.

SCHEDULE AND ROLES

Five days, two builders

Draft is Wednesday night; final is Friday night.



Working plan (Sunday: agree the idea and roles). Dates are from the event schedule; confirm the deadline timezone on the dashboard.

Day	Goal	Done when
Mon 21	Hamiltonian, clean map checked against the reference lines; time one noisy point and fix N and grid; game skeleton (fog, pings, painting)	Clean diagram looks right; noisy runs started; clickable prototype
Tue 22	Full noisy dataset; scoring wired to the real data; baseline agent	Game runs on real data end to end
Wed 23 (draft)	Three phase diagrams; first noise analysis; levels 1-3 playable; 1-page summary	Draft submitted before the deadline
Thu 24	Pings-vs-accuracy study; boss level; share card; 5+ playtests; notebook cleanup	Numbers for the writeup; freeze game features at noon
Fri 25 (final)	Freeze code by the afternoon; 2-3 page writeup; slides; rehearse twice; submit early	Submission uploaded with time to spare

Who	Owns
Tanish	Decisions; physics understanding and review of every figure; presentation and demo; final say on claims
Claude	Builds the simulation, dataset, classifier, game, agent, experiments and figures; drafts the writeup for Tanish to edit
Teammates	Review this plan; playtest Thu (target five people, no coaching); read the writeup for clarity; rehearse the demo as judges

If we fall behind, cut in this order: (1) Level 5 finale, (2) share card and daily map, (3) boss level polish, (4) optional error mitigation and QCNN. **Never cut:** the three phase diagrams, the noise analysis, the notebook, the writeup.

What we show, what we submit, what could go wrong

Presentation length in the handout is 5-7 minutes. Proposed script:

Time	Beat
0:00-0:45	Hook: a judge takes the laptop. 'Find the four phases in 20 pings.'
0:45-1:45	The physics on one slide: the real reference diagram and the three-line explanation
1:45-3:15	Live play: judge vs the agent on level 2; reveal and score
3:15-4:45	Noise results: three phase diagrams, measured boundary shifts, which phase shrinks first and why
4:45-5:45	Pings-vs-accuracy study (only what we actually measured)
5:45-6:30	Limits (small N, floating phase) and what we would do next

Required deliverable	Where it comes from
Notebook (.ipynb)	Simulation, classifier, diagrams, experiment; organized and commented
Phase diagrams $p = 0, 0.01, 0.05$	Pipeline steps 1-2; the same code feeds the game
Writeup (2-3 pages)	Method, results, physical interpretation; structure follows the level ladder
Presentation (5-7 min)	Script above

Risk	Mitigation
Noisy simulation is too slow	Time one point Monday; drop to $N = 6$ and a 16×16 grid if needed
Floating phase is not resolved at this size	Expected per the handout; the boss level teaches this and we report it honestly rather than claim detection
Game eats science time	Science tasks scheduled first each day; game features freeze Thursday noon
Agent does not beat the baselines	Report it as a finding; the study is still a valid noise-analysis tool
Finite-size lines are only qualitative	Say so on the slide; compare trends and boundary shifts, not exact values
Deadline timezone	Confirm on the dashboard; submit at least a few hours early

Decisions to agree today, and questions to be ready for

Decisions we need from you

Decision	Default if nobody objects
Are we entered in the Scientific Track (Open Challenge)?	Yes; this plan assumes it
Game platform	Single-file browser game, shareable link
System sizes	Clean $N = 8$, noisy $N = 6$; revisit after Monday's timing
Playtesters on Thursday	Teammates plus a few friends, five or more
Who presents	Tanish, with a teammate running the laptop

Questions judges may ask (answers we should own)

Question	Short answer
Is the game running a real quantum computer?	No. It samples from precomputed simulation results (PennyLane / exact diagonalization) with real shot noise. The simulation, not the game, produces the required diagrams.
Why only 6-8 spins?	Simulation cost. The handout notes finite-size limits; we compare trends, and treat $N \geq 12$ as a bonus if time allows.
Did you detect the floating phase?	We report what the data shows. The handout calls it very hard at small size; our boss level makes that difficulty visible.
What is new here?	The measurement-budget study: how many pings a boundary needs under noise, by strategy. Framed as our experiment, not a general result.
Why does noise blur the phases?	Ordered phases are defined by correlations between spins; gate errors scramble them, pushing ordered regions toward the paramagnet. The antiphase's more complex pattern should be more fragile; we will test that rather than assume it.

Sources

Quantum Coalition QSITE 2026 challenge repository: github.com/benmcdonough20/QSITE-2026-QuantumCoalition (root README, Scientific Track README and starter kit: reference boundaries, noise model, rubric, deliverables). PennyLane demo: ANNNI Phase Detection (pennylane.ai/qml/demos/tutorial_annni). PennyLane challenge: A Noisy Heisenberg Model (pennylane.ai/challenges/heisenberg_model). Event dates from the Q-Site event schedule screenshot supplied by Tanish. Real-data figures: exact diagonalization written for this plan (numpy), 8 spins, periodic boundary conditions; the game screens, level graphics and pipeline are mock-ups.